GreenBytes AI

Intelligent Storage Optimization to Reduce Digital Carbon Footprints

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Virtual Internship | National Internship Portal
In Collaboration with **IBM SkillsBuild & AICTE**

Primary Target: SDG 13 (Climate Action)

The Digital Waste Crisis

Analyzing how unstructured raw data stored on cloud systems acts as a key driver of modern energy consumption and scope 2 emissions.

Redundant Data, Real Emissions

The Dark Data Problem

Up to **55% of all stored data** is categorized as 'dark data'—unreferenced database assets, outdated system logs, and redundant files stored endlessly.

Keeping these unused files active requires continuous disk operations, power distribution, and refrigeration 24 hours a day, adding heavy loads to local energy grids.

The Operational Gap

Traditional rule-based file deletion scripts lack contextual intelligence. They look only at names, occasionally triggering accidental deletions of critical logs.

System administrators cannot manually audit millions of files safely. There is an urgent need for semantic and highly scalable contextual file reasoning.

GreenBytes Solution Architecture

Leveraging IBM Granite Models

GreenBytes AI works as an automated storage scanning daemon. It crawls directories and processes file headers contextually instead of superficially.

- **Metadata Extraction:** Pulls timestamps, author logs, and text summaries safely.
- **Semantic Analysis:** IBM Granite analyzes content to flag redundancy risk score.
- **Emissions Mapping:** Estimates saved active server wattage and converts it into carbon offsets.

Intelligent De-duplication Flow

1. Directory Scan

Python script scans targeted storage clusters for file patterns.

2. Metadata Parsing

Extracts content chunks, file properties, and active records safely.

3. Granite Evaluation

IBM Granite models analyze summaries to score archive redundancy.

4. BOB Orchestration

IBM BOB validates the decision workflow before suggesting actions.

Responsible AI Guardrails

- **Privacy & Security First:** GreenBytes processes file contents locally or via encrypted channels. No customer personal identifiable information (PII) is cached or exposed during processing.
- **Human-in-the-Loop Authority:** The AI acts strictly as an analytical advisor. It generates clean reports and options, leaving actual deletion controls 100% in human administrator hands.
- **Transparent Audit Trails:** Every recommendation is backed by a short, human-readable rationale explanation block generated by the IBM Granite LLM.
- **Verifiable Metrics:** Storage savings are converted to localized carbon offsets (CO_2 reductions) using validated public grid parameters for extreme auditability.

Projected Carbon Offset Gains

Using deep semantic processing to safely classify complex log archives unlocks massive storage capacity gains, translating to direct energy savings in global cloud centers.

Join the Green Computing Revolution

Sustainable development meets responsible Artificial Intelligence.

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